

🎓 3-Week Prep Plan for LLMs & Prompt Engineering (Before OpenAI Agent SDK)

✨ Objective:

Prepare students who have completed Python programming to understand AI vs Generative AI, learn LLM fundamentals, explore prompt engineering techniques, and gain hands-on practice across platforms like ChatGPT, Gemini, Grok, and DeepSeek.

📖 Week 1: AI & LLM Fundamentals

🎯 Goals:

- Understand the difference between Traditional AI and Generative AI
- Learn how LLMs (Large Language Models) work

Daily Plan:

Day 1-2: 🤖 What is AI vs Generative AI

- 📌 Task: Write a 200-word comparison between rule-based AI and Generative AI
- 🔗 Resource: [LearnPrompting: Introduction](#)

Day 3: 🌐 Generative AI in Practice

- 📌 Task: List 3 traditional AI use cases vs 3 generative AI examples

Day 4-5: 🧠 How LLMs Work

- 📌 Task: Summarize how transformers and attention mechanisms work in LLMs

Day 6-7: 📊 Strengths & Weaknesses of LLMs

- 📌 Task: Identify 3 strengths and 3 limitations with examples

Week 2: Prompt Engineering Techniques

Goals:

- Learn core prompting styles (zero-shot, few-shot, chain-of-thought)
- Practice across multiple LLM platforms

Daily Plan:

Day 1: Zero-shot & Few-shot Prompting

-  Prompt Example:

Zero-shot: "Explain osmosis in one paragraph."

Few-shot:

Example 1: "The cat sat on the mat." → "The feline rested on the rug."

Now rephrase: "The sun kissed the ocean."

Day 2: Chain-of-Thought Prompting

-  Prompt Example:

"If A is older than B, and B is older than C, who is youngest? Think step-by-step."

Day 3: Role Prompting

-  Prompt Example:

"Act as a history tutor and explain the Cold War to a 12-year-old."

Day 4: Output Formatting

-  Prompt Example:

"List 3 benefits of renewable energy in bullets."

"Return JSON with keys: topic, summary, benefits."

Day 5: Meta-Prompting

- 💡 Prompt Example:

"Here is my prompt: [your prompt]. How can I improve it?"

Day 6–7: Cross-Model Practice

- 📌 Task: Run the same prompt on ChatGPT, Gemini, Grok, DeepSeek
- 📄 Log outputs, compare clarity, structure, and correctness



Week 3: Deep Dive + Tool Thinking

🎯 Goals:

- Develop intuition for LLM behavior
- Apply prompts to real-world scenarios
- Sketch a personal LLM-powered tool idea

Daily Plan:

Day 1: Transformer Internals

- 📌 Task: Write a 150-word explanation of self-attention in transformers

Day 2: Strengths vs Weaknesses

- 📌 Task: Write 3 excellent and 3 poor use cases of LLMs

Day 3: Prompt Style Comparison

- 📌 Task: Write and test a zero-shot, few-shot, and instruct prompt for the same task

Day 4: Advanced Prompt Techniques

- 📌 Task: Create a persona + reasoning prompt and iterate using meta feedback

Day 5: Prompt Refinement Practice

- 📌 Task: Refine a weak prompt with clarity, formatting, and context

Day 6: Sketch a Tool

- 📌 Task: Write a simple idea for an LLM assistant (e.g. FAQ bot or calendar planner)

Day 7: Reflection & Showcase

- 📌 Task: Submit 1-page reflection with prompt examples and model output comparisons

✅ End-of-Break Deliverables:

- 📄 AI vs GenAI comparison write-up
- 📄 150-word LLM mechanism summary
- 📋 Prompt table (5 prompts tested on ≥ 2 platforms)
- 💡 LLM tool idea sketch
- 📝 Reflection page

🌐 Platforms to Explore:

- ChatGPT
- Gemini (Google)
- Grok (xAI)
- DeepSeek

📚 Learning Resources:

- [LearnPrompting.org - Complete Guide](https://learnprompting.org)
- [Hugging Face LLM Course](https://huggingface.co/course)
- [OpenAI Prompting Guide](https://openai.com/prompting)
- [Wikipedia: Generative AI](https://en.wikipedia.org/wiki/Generative_artificial_intelligence)
- [Medium: Attention Mechanism in LLMs](https://medium.com/@josh123456789/attention-mechanism-in-llms)